

### Question 1

What is the definition of a telecommunications system, and what is its primary scope in modern communication?

**A telecommunications system is a group of devices and technologies that help people send and receive information over long distances. An example, everyday when we make a phone call or even send a text, this is considered a telecommunications system.**

### Question 2

Explain the basic components of a communication system and describe the role of each: transmitter, channel, and receiver.

**The transmitter converts information into a signal for transmission. Channel is when the signal then travels from the transmitter to the receiver. After that the Receiver accepts the signal which then starts to reconstruct the original information.**

### Question 3

Differentiate between analog and digital communication systems. What are the key performance metrics used to evaluate them (e.g., bandwidth, noise, latency)?

**Analog communication uses signals that are continuous and more likely to be affected by noise, whereas digital communication uses binary data instead making it clearer and better for communications.**

**Some key performance metrics are bandwidth, noise and latency. Bandwidth measures how much data can be transmitted, noise refers to unwanted interferences while being transmitted, and latency on the delay.**

### Question 4

Describe the main types of communication (voice, data, video, multimedia) and explain how network topology and system architecture influence their transmission.

**Voice communication is used for phone calls, data communication commonly**

**found used in emails, video communication used for streaming or using the camera and lastly multimedia communication combine the use of video, voice, images, etc.**

### **Question 5**

How do different network topologies (such as star, mesh, ring, and bus) impact the performance, reliability, and scalability of a communication system?

**A star topology is easy to manage as it has all devices connected, mesh topology has data travel through multiple paths in case connections fail, ring topology allows data to move in an organized manner and bus topology is simple to install but easily make the entire network stop working if main cable fails.**